02_analytics_engineer

Principal Analytics Engineer Review

Headline kill. The entire "85% root-cause accuracy" proof is a closed loop: the eval injector perturbs *aggregates* and then computes the "true" mix/rate/interaction split "by **the same decomposition algebra Helios is graded on**" (Bible §20.2, step 3), so the benchmark grades Helios's arithmetic against Helios's own arithmetic applied to data Helios itself constructed. That is a unit test of an algebraic identity (`ΔR = mix + rate + interaction` , which is *true by definition*), dressed up as an empirical accuracy claim. Worse, the input to that decomposition — `fct_funnel_by_dim` — is documented to hold exactly **one dimension at a time** (DATA_MODEL.md §5.10), yet 6 of the 50 labeled scenarios target two-dimensional cells like `Paid Search × mobile` (scenarios.yaml). The mart physically cannot represent the segment the label says is the root cause. The headline number is unearned twice over.

[CRITICAL] The decomposition input mart cannot represent the segments the eval grades against

Claim attacked. `fct_funnel_by_dim` is "the funnel rolled up by a single **canonical dimension** ... one canonical dimension at a time" with PK `(date_key, dimension, dimension_value)` , and it is "the direct input to the mix-vs-rate decomposition (`decompose_change`)" (DATA_MODEL.md §5.10, §2.1 catalog). Bible §20.2 says the decomposition gold target is computed on the `fct_daily_funnel` grain. **Why it fails.** `decompose_change` requires a `(w_i, r_i)` vector over the *segments of one partition of the population*. Six labeled scenarios (S001 and five siblings) declare `root_cause_segment: {channel_group: "Paid Search", device_category: "mobile"}` — a cross-tab cell. `fct_funnel_by_dim` long-unpivots to one dimension, so it has a `channel_group=Paid Search` row and a `device_category=mobile` row but **no `Paid Search × mobile` row**; their session counts overlap and cannot be summed or differenced into a single weight. To grade these scenarios you must decompose over the `channel_group × device_category` cross product, which only `fct_daily_funnel` carries (and only for its four hardcoded dims). So the two decomposition feeds disagree on grain, and the single-dimension mart the doc names as "the direct input" is the wrong table for 12% of the benchmark. This is a design contradiction, not a typo. **Fundamental.** The decomposition input grain contradicts the labels.

[CRITICAL] "Diagnose WHY" is arithmetic attribution, not causal inference

Claim attacked. The thesis: an engine that "diagnoses **why** an e-commerce funnel moved" (CLAUDE.md §1); the decomposition "dissolves Simpson's paradox" and is "the technical centerpiece" (Bible §19, §4). **Why it fails.** `ΔR = Σ Δw_i·r_i(t0) + Σ w_i(t0)·Δr_i + Σ Δw_i·Δr_i` is a tautological re-expression of an aggregate change — it tells you *where in the segment grid* the number moved and whether the mover was composition or in-segment rate. It says nothing about *cause*: a deploy, a price change, a competitor, a promo, a holiday, a tracking break. The mart layer has no event log, no deploy calendar, no price-change feed except a single SCD2 snapshot on item price (`snap_dim_items`), and explicitly **no cost data** (DATA_MODEL.md §5.3). So the strongest honest claim is "the rate of checkout-to-purchase fell *in the Paid Search mobile cell*," which the system then narrates as a *why*. Relabeling "where" as "why" is the central overclaim, and grounding the SQL does nothing to prevent the wrong causal story over correct numbers. **Fundamental.** Attribution algebra is not causation.

[HIGH] The whole channel decomposition rests on session-scoped source/medium that GA4's obfuscated sample barely populates

Claim attacked. "Prefer the session-scoped `event_params.source` / `medium` … fall back to the user-level `traffic_source` struct only when the session params are NULL" (DATA_MODEL.md §5.3); channel is "the precondition for honest mix-vs-rate decomposition" (§1.3). **Why it fails.** In the `ga4_obfuscated_sample_ecommerce` export, per-event `event_params.source` / `medium` are sparse and frequently absent outside the acquisition event; `traffic_source.*` (user first-touch) is the field that is reliably populated. The doc itself warns first-touch fallback "mis-attributes every returning session to its original acquisition channel" (§5.3) — yet the coalesce makes first-touch the fallback precisely when session scope is NULL, which on this dataset is *most non-landing events*. So `channel_group` , the headline decomposition dimension, silently degrades toward first-touch for a large share of sessions, and `int_ga4__sessionized` resolves it with `max(event_source)` / `max(event_medium)` (DATA_MODEL.md §5.5 SQL) — a non-deterministic pick of "a" source within the session, not the earliest. Two different sources in one session resolve by alphabetical MAX, not by first-touch order. The channel axis the entire mix-vs-rate story stands on is itself a noisy, partially-first-touch, order-insensitive guess. **Fundamental.** The dataset cannot supply clean session-scoped channel.

[HIGH] The cohort SQL is broken and the retention numerators are duplicated across ages

Claim attacked. `fct_cohorts` "pre-computing `cohort_size` and `retained_users_d1/d7/d30` " so retention metrics "resolve as simple `SUM(retained_users_dN)/SUM(cohort_size)` " (DATA_MODEL.md §6.3, with copy-usable SQL). **Why it fails.** The shipped query does not compile and would not mean what it claims if it did. It writes `cross join unnest([1,7,30]) as age_days with offset` but then selects `ages.age_days` and groups by `ages.age_days` — the table alias is `age_days`, there is no `ages`, and `with offset` introduces a second unnamed column; this is a reference error. Worse, the three numerators are `date_diff(...) between 1 and 1`, `between 1 and 7`, `between 1 and 30` — *hardcoded constants that ignore the `age_days` axis entirely*. So every one of the three age rows for a cohort carries the **same** d1/d7/d30 triple; the `age_days` grain is decorative. And the `returns` CTE computes a per-user `min(...) over (partition by user_pseudo_id) as ignore_me` window it never uses, then joins on `user_key` against a fan-out of every session date. This is not "production-grade" SQL (§Purpose); it is pseudocode that was never run — consistent with the project's own status: no code built yet. **Fixable.** Rewrite the snippet — but it discredits "production-grade."

[HIGH] day_30 retention is mostly unobservable on a 3-month frozen sample — the metric is theater

Claim attacked. `day_30_retention` is a first-class semantic metric (semantic_layer.yaml) and a named eval grain; the doc concedes "the ~3-month window right-censors `day_30` for late cohorts" and says it "must be restricted to *mature* cohorts" (DATA_MODEL.md §3 catalog, §6.3). **Why it fails.** The window is 2020-11-01 to 2021-01-31, 92 days. A cohort needs ≥30 days of forward observation for an *uncensored* d30, so only acquisition weeks in roughly Nov 1–Jan 1 qualify, and the *useful, mature* cohorts collapse to a handful of early weeks — exactly the weeks with the least traffic at the dataset's leading edge. After the "restrict to mature cohorts + minimum cohort_size" filters the doc mandates, `day_30_retention` is computed from a tiny, non-representative slice. Shipping it as a governed headline metric on a static historical export is freshness/maturation theater: it can never mature further because the data never advances. The metric exists to look complete, not to be trustworthy. **Fundamental.** A frozen 92-day window cannot support d30 cohorts.

[HIGH] Source-freshness and the autonomous schedule are pure ceremony on a static export

Claim attacked. "`dbt source freshness` runs first and its exit code gates everything downstream ... A stale source must block diagnosis" (DBT_GUIDE.md §2); the product's "heartbeat is the autonomous scheduled run" (CLAUDE.md §1). **Why it fails.** DBT_GUIDE.md §2 then admits the sample is static, the newest shard is `events_20210131`, "Real freshness against 'now' (2026) is therefore meaningless: every freshness check would scream `error`," so the gate is disabled on the sample and `dbt_utils.recency` is downgraded to `warn`. So on the only data that exists, the freshness machinery is wired to do nothing. Every "scheduled run" sees the identical frozen 92 days; there is no new shard to detect, no anomaly that wasn't there yesterday, no day_30 that matured, nothing to monitor. "Always-on autonomous diagnosis" over a static table is a cron job that recomputes the same answer forever. The freshness contract is "real, production-grade, and fully implemented" only in the sense that it is correctly configured to be inert. **Fundamental.** Nothing live exists to monitor or refresh.

[MEDIUM] The 47-metric semantic layer is bloated with metrics this dataset cannot honestly support

Claim attacked. "47 governed metrics" as a feature (DATA_MODEL.md §1, §1.4); 13 are category `supporting` and 4 are `acquisition` (semantic_layer.yaml category counts). **Why it fails.** `cac_proxy` is labeled "CAC Proxy (**NOT** true CAC)" and its own caveats say "the GA4 export has **no** cost/spend column, so a dollar CAC is impossible here" — it is `paid_sessions/new_purchasers`, a "unitless ratio" with a five-bullet do-not-use list. A metric that requires four paragraphs explaining what it *isn't* is liability, not asset. The product cluster (`product_view_rate`, `product_conversion_rate`, `item_view_sessions`, etc.) depends on item-level `view_item` array tracking that the doc itself says cannot be linked to purchases within a session or even a cookie (DATA_MODEL.md §7.5), so "product conversion" is flagged as a cross-session approximation the Critic must caveat — i.e. it isn't a conversion rate. Thirteen `supporting` measures (`retained_users_d1/d7/d30`, `item_views`, `orders_with_item`, `paid_sessions` ...) are mostly denominators/numerators of the headline metrics re-exposed as first-class metrics, inflating the count. The 47 is a surface-area number; the load-bearing set is perhaps 15. **Fixable.** Cut the proxies and the duplicate components.

[MEDIUM] "Reconcile to the cent" and "0 dangling refs" are asserted, not earned — and partly self-contradicting

Claim attacked. Revenue must "match the source to the cent (`revenue_reconciles` test)" (CLAUDE.md §8 keystone 4); `SUM(session_revenue)` over `fct_funnel` "equals `SUM(gross_revenue)` over `fct_orders` at the grand total (within the 0.5% `reconcile` tolerance)" (DATA_MODEL.md §8.4); the YAML "passed a referential-integrity LINT." **Why it fails.** "To the cent" and "within 0.5%" are not the same claim, and they appear in the same project as co-equal guarantees. More importantly, the two revenue paths are not built the same way: `fct_orders` derives `gross_revenue` from `ANY_VALUE(ecommerce.purchase_revenue_in_usd)` after grouping on `(order_key, session_key, user_pseudo_id, order_ts)` (DATA_MODEL.md §8.2 SQL) — but if a `transaction_id` 's duplicate rows carry *different* timestamps, that GROUP BY produces **two rows for one transaction**, defeating the dedup the section is named after. Meanwhile `session_revenue` dedups on `transaction_id` within the session. A purchase whose duplicate rows split across two `ga_session_id` s (e.g. a midnight-crossing session, which the doc says splits, §5.1) lands in two `fct_funnel` rows but one `fct_orders` row — so the grand totals are *not guaranteed* equal even in principle. The reconciliation is asserted as an invariant; the dedup logic shown does not establish it. And a YAML lint proves names resolve, not that any number reconciles to anything — no query has run. **Fixable.** Fix the dedup keys; but the "to the cent" claim is unproven.

[MEDIUM] The wide-fact assumption contradicts the documented marts the decomposition needs

Claim attacked. "Wide facts … descriptive dimensions (`device_category`, `country`, `channel_group`, `is_new_user`, `landing_page`) are denormalized directly onto `fct_funnel`" so the layer "slices a metric by simply `GROUP BY`-ing a column on the same fact, with no join" (DATA_MODEL.md §1.5). **Why it fails.** The `fct_funnel` column list in §5.8 carries only `device_category`, `country`, `is_new_user` (plus keys and `reached_*`). It does **not** carry `operating_system`, `browser`, `region`, `source`, `medium`, `campaign`, `landing_page`, or `session_number_bucket` — yet the `revenue`, `sessions`, `users`, and `revenue_per_session` metrics in semantic_layer.yaml all advertise `dimensions_supported` including `operating_system, browser, region, source, medium, campaign, landing_page, session_number_bucket`. So the semantic layer promises to slice `revenue` by `landing_page` on a fact that does not contain `landing_page`. Either the "no join at query time" property is false (you must join back to `fct_sessions`, which *does* carry those dims) or the fact must be widened far beyond what §5.8 documents. The single-source-of-truth catalog and the registry disagree on what columns exist — and the agents will request dims that aren't there. **Fixable.** Widen the fact or document the joins — but as written they conflict.

[MEDIUM] The single-source-of-truth discipline is already violated in the canonical docs

Claim attacked. "metric/dimension definitions live **only** in `models/semantic/semantic_models.yml`" and "`semantic-mcp`'s `registry:` … must point at that exact path" (CLAUDE.md §5, §8 keystone 2); the Bible's Reference Card "wins over any prose anywhere" (CLAUDE.md §2). **Why it fails.** The actual registry is `models/semantic/semantic_layer.yaml` (v2, 47 metrics), whose header explicitly "Supersedes `semantic_models.yml` (v1)." Yet CLAUDE.md §5 *and* §8, plus MCP_ARCHITECTURE.md §lines 57/74/159, all still name `semantic_models.yml` as the keystone and the registry path. The playbook itself logs this as a live bug ("Filename drift — `mcp_servers.yaml` and `hallucination.py` must point at `semantic_layer.yaml` not `semantic_models.yml`; a stale path makes the AST check pass everything," 05_m10_m12.md). So the document that defines the *single source of truth rule* points the hallucination gate at a file that doesn't exist — meaning the "0 hallucinated columns" check would load nothing and pass everything. Separately, the registry's own resolver contract — "maps `metric_name` → `name`, `sql_definition` → `sql`, `aggregation_method` → `agg`" (semantic_layer.yaml header) — is an adapter the schema describes but no code implements; `build_query` per the Bible expects `name` / `sql`. A project whose three "resume-point" docs disagree about the name and shape of its governance keystone has not earned the word "governed." **Fixable.** Rename and align — but the discipline it claims is already broken.

[MEDIUM] `engaged_session` and source resolution use `MAX` / `ANY_VALUE` as if attributes were stable, silently corrupting slices

Claim attacked. Device/geo are "`any_value` within the session (stable per visit)"; engagement is `max( … )`, source is `coalesce(max(event_source), max(first_touch_source))` (DATA_MODEL.md §5.2, §5.5 SQL). **Why it fails.** `ANY_VALUE` and `MAX` are not "the session's value" — they are *whatever the engine returns*, and the doc conflates the two. `MAX(event_source)` over a session with both `google` and `(direct)` returns `google` by string ordering, not by recency or first-touch; `MAX(device_category)` over a session that GA4 logged as both `mobile` and `tablet` (device switches, app→web) silently picks `tablet`. These columns are the *exact* slicing keys the decomposition partitions on. When the eval injects a clean `rate_multiplier: 0.55` into the `Paid Search × mobile` cell, but real sessionization smears a fraction of those sessions into `tablet` or `(direct)` via MAX, the injected ground-truth cell and the reconstructed cell don't line up — and the MAPE/accuracy the system reports against its own labels degrades for reasons that have nothing to do with diagnosis quality. The keystone the doc says "fails silently" indeed does, by its own aggregation choices. **Fixable.** Use ordered `ARRAY_AGG … LIMIT 1` consistently — but as written the spine is non-deterministic.